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**DATASET TOPIC: PHARMA-HEALTHCARE
SALES DATA**

**TITLE: HEALTHCARE SALES
ANALYSIS**

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DATASET: <https://github.com/sssingh/pharmaceutical-sales-analysis-powerbi?tab=readme-ov-file#dataset>

Data Cleaning using Power Query Editor

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- 28. All transformations completed by using Power BI Power Query Editor.**
- 29. Close and Apply to load the data into Power BI.**

DAX and Visualization

DAX (Data Analysis Expressions)

1. Created a New Calculated Column Channel Type in Pharma Sales Data
2. Calculated Measures - Total Number of Sales
3. Created Calculated Table “Sales Details by Country”
4. Calculated Measure - Average Sales Amount
5. Calculated Measure - Total Sales Amount
6. Created a New Calculated Column Sales Team Details

Data Modeling

Data Visualization

1. Sales by Year and Channel
2. Sales by Country
3. Sales by Month and Year
4. Sales by Sub-Channel
5. Sales by Sales Team and Product Class
6. Sales by Managers
7. Cards - Total Sales Amount
8. Cards - Number of Sales
9. Slicer - Country
10. Slicer - Distributors
11. Slicer - Product Class

Data Cleaning using Power Query Editor

1. Data Imported

- Imported “pharma-data.csv” file into Power BI.
- Select “Based on Entire Dataset” under Data type detection.
- Click on “Transform Data” to open the table in Power BI Power Query Editor.

The screenshot shows the Power BI Desktop interface with the 'pharma-data.csv' import dialog box open. The dialog box displays the following settings:

- File Origin: 65001: Unicode (UTF-8)
- Delimiter: Comma
- Data Type Detection: Based on entire dataset

The data table is as follows:

Distributor	Customer Name	City	Country	Latitude	Longitude	Channel
Gottlieb-Cruickshank	Zieme, Doyle and Kunze	Lublin	Poland	51.2333	22.5667	Hospital
Gottlieb-Cruickshank	Feest PLC	Świecie	Poland	53.4167	18.4333	Pharmacy
Gottlieb-Cruickshank	Medhurst-Beer Pharmaceutical Limited	Rybnik	Poland	50.0833	18.5	Pharmacy
Gottlieb-Cruickshank	Barton Ltd Pharma Plc	Czeladź	Poland	50.3333	19.0833	Hospital
Gottlieb-Cruickshank	Keeling LLC Pharmacy	Olsztyn	Poland	53.78	20.4942	Pharmacy
Gottlieb-Cruickshank	Runte-Marquardt Pharmaceutical Ltd	Olecko	Poland	54.0333	22.5	Hospital
Gottlieb-Cruickshank	Blick, Pacocha and Schowalter	Inowrocław	Poland	52.7958	18.2611	Pharmacy
Gottlieb-Cruickshank	Leuschke PLC Pharmacy	Ciechanów	Poland	52.8817	20.6106	Pharmacy
Gottlieb-Cruickshank	Miller-Satterfield Pharma Plc	Nidzica	Poland	53.3583	20.425	Hospital
Gottlieb-Cruickshank	Bashirian-Kassulke Pharma Plc	Kraków	Poland	50.0614	19.9372	Hospital
Gottlieb-Cruickshank	Wolff Group Pharm	Gubin	Poland	51.95	14.7333	Hospital
Gottlieb-Cruickshank	Denesik, Walter and Beatty Pharma Plc	Ostrów Mazowiecka	Poland	52.8	21.9	Pharmacy
Gottlieb-Cruickshank	Breitenberg-Kuhn Pharmacy	Radom	Poland	51.4036	21.1567	Hospital
Gottlieb-Cruickshank	McClure, Zemlak and Dibbert Pharma Plc	Gdańsk	Poland	54.3667	18.6333	Pharmacy
Gottlieb-Cruickshank	Block-Romaguera Pharmaceutical Limited	Tarnowskie Góry	Poland	50.45	18.8667	Hospital
Gottlieb-Cruickshank	Hahn, Ruthford and Gislason Pharmaceutical Limited	Łaziska Górne	Poland	50.15	18.8333	Hospital
Gottlieb-Cruickshank	D'Amore and Sons	Wągrowiec	Poland	52.8	17.2	Hospital
Gottlieb-Cruickshank	Koss Ltd Pharmaceutical Limited	Lwówek Śląski	Poland	51.1167	15.5833	Hospital
Gottlieb-Cruickshank	Fadel-West	Złotoryja	Poland	51.1264	15.9198	Pharmacy
Gottlieb-Cruickshank	Lindgren-Simonis Pharm	Biskupiec	Poland	53.8647	20.9569	Pharmacy
Gottlieb-Cruickshank	Spinka, Schowalter and Conn Pharmaceutical Limited	Bilgoraj	Poland	50.55	22.7333	Pharmacy

The 'Transform Data' button is highlighted in green.

2. Data Transformed into Power Query Editor

- Transformed the Data Table into “Power Query Editor” to complete Data Cleaning and Transformation steps.

The screenshot displays the Power Query Editor window titled "Untitled - Power Query Editor". The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing options like Close & Apply, New Source, Recent, Enter Data, Data source settings, Manage Parameters, Refresh Preview, Properties, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Remove Rows, Sort, Split Column, Group By, Data Type, Use First Row as Headers, Replace Values, Merge Queries, Append Queries, Combine Files, Text Analytics, Vision, and Azure Machine Learning.

The main area shows a data table with the following columns: **Distributor**, **Customer Name**, **City**, and **Country**. The table contains 25 rows of data, all from the distributor "Gottlieb-Cruickshank". The formula bar at the top indicates the query is defined by the formula: `= Table.TransformColumnTypes(#"Use First Row as Headers", {"Distributor", type text}, {"Customer`.

On the right side, the "Query Settings" pane is open, showing the "Properties" tab with the name "pharma-data" and the "Applied Steps" tab with the steps "Source", "Use First Row as Headers", and "Change Type".

At the bottom, the status bar indicates "18 COLUMNS, 999+ ROWS" and "Column profiling based on top 1000 rows". The bottom right corner shows "PREVIEW DOWNLOADED AT 04:57 P".

3. Changed Data Type

- Changed Data Type of Columns.
 - Latitude, Longitude, Quantity - Text into “Number”.
 - Price, Sales - Text into “Fixed Decimal Number” (Currency).
 - Year - Text to “Data and Time”.

```
= Table.TransformColumnTypes("#Use First Row as Headers",{{"Distributor", type text}, {"Customer Name", type text}, {"City", type text}, {"Country", type text}, {"Latitude", type number}, {"Longitude", type number}, {"Channel", type text}, {"Sub-channel", type text}, {"Product Name", type text}, {"Product Class", type text}, {"Quantity", Int64.Type}, {"Price", Currency.Type}, {"Sales", Int64.Type}, {"Month", type text}, {"Year", Int64.Type}, {"Name of Sales Rep", type text}, {"Manager", type text}, {"Sales Team", type text}})
```

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with the following columns: Quantity, Price, Sales, Month, and Year. The table contains 21 rows of data. A tooltip "Changed Data Type of Quantity, Price, Sales and Year" is visible over the table. The right sidebar shows the "Query Settings" for the "pharma-data" query. Under "APPLIED STEPS", the steps listed are "Source", "Use First Row as Headers", and "Changed Data Type of Quanti...". The bottom status bar indicates "18 COLUMNS, 999+ ROWS" and "Column profiling based on top 1000 rows".

4. Formatted the Sales Team Column into Uppercase

- Formatted the “Sales Team” Column from Propercase into “UPPERCASE”, to Highlight the “Sales Team”.

The screenshot displays the Power BI Desktop interface with the 'pharma-data' query selected. The 'Sales Team' column is highlighted in the data table. The 'Transform' ribbon is active, showing the 'Format' button in the 'Text Column' group. A tooltip for the 'Format' button indicates the action: 'Transformed Format of Sales Team into UPPERCASE'. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Transformed Format of Sales ...'. The data table shows columns: Month, Year, Name of Sales Rep, Manager, and Sales Team. The 'Sales Team' column values are being transformed from proper case (e.g., DELTA, BRAVO) to uppercase (e.g., DELTA, BRAVO).

	Month	Year	Name of Sales Rep	Manager	Sales Team
1	January	2018	Mary Gerrard	Britanny Bold	DELTA
2	January	2018	Jessica Smith	Britanny Bold	DELTA
3	January	2018	Steve Pepple	Tracy Banks	BRAVO
4	January	2018	Mary Gerrard	Britanny Bold	DELTA
5	January	2018	Anne Wu	Britanny Bold	DELTA
6	January	2018	Thompson Crawford	James Goodwill	ALFA
7	January	2018	Sheila Stones	Britanny Bold	DELTA
8	January	2018	Mary Gerrard	Britanny Bold	DELTA
9	January	2018	Stella Given	Alisha Cordwell	CHARLIE
10	January	2018	Morris Garcia	Tracy Banks	BRAVO
11	January	2018	Erica Jones	James Goodwill	ALFA
12	January	2018	Abigail Thompson	Tracy Banks	BRAVO
13	January	2018	Abigail Thompson	Tracy Banks	BRAVO
14	January	2018	Jessica Smith	Britanny Bold	DELTA
15	January	2018	Anne Wu	Britanny Bold	DELTA
16	January	2018	Morris Garcia	Tracy Banks	BRAVO
17	January	2018	Daniel Gates	Alisha Cordwell	CHARLIE
18	January	2018	Morris Garcia	Tracy Banks	BRAVO
19	January	2018	Anne Wu	Britanny Bold	DELTA
20	January	2018	Alan Ray	James Goodwill	ALFA
21	January	2018	Timothy Gray	Alisha Cordwell	CHARLIE

18 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 05:38 PM

5. Checked Distinctive Values of All Column, to find out missing values or blank cells Using Filter Option

- Checked Distinctive values of all Column, to find out the blank and null values using “Filter” option.
- Checked for Duplicate Values using “Filter and Sort” option.
- There is no Duplicate and Missing Values.

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a table named 'Sales' with columns: Sales, Month, Year, Name of Sales Rep, and Manager. A context menu is open over the 'Sales' column, showing options like 'Sort Ascending', 'Sort Descending', 'Clear Sort', 'Clear Filter', 'Remove Empty', and 'Number Filters'. The 'Number Filters' option is selected, and a filter pane is visible on the right side of the table. This pane shows a search bar and a list of years: 2017, 2018, 2019, and 2020, all of which are checked. The status bar at the bottom indicates '18 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Query Settings

Query Name: pharma-data

APPLIED STEPS

- Source
- Use First Row as Headers
- Changed Data Type of Quanti...
- Transformed Format of Sales ...
- Checked Distinctive Values of ...

6. Checked for Missing Values Using Column Quality

- Again Checked Missing Values and error values Using “Column Quality” Option.

The screenshot displays the Power BI Desktop interface with the 'Column Quality' tool applied to the 'pharma-data' table. The tool's output shows the following quality metrics for each column:

Column	Valid	Error	Empty
Country	100%	0%	0%
Latitude	100%	0%	0%
Longitude	100%	0%	0%
Channel	100%	0%	0%
Sub-channel	100%	0%	0%

The 'Applied Steps' pane on the right shows the following steps:

- Source
- Use First Row as Headers
- Changed Data Type of Quanti...
- Transformed Format of Sales ...
- Checked Distinctive Values of ...

7. Rounded Off Latitude Column

- Rounded Off “Latitude” Column with 2 Decimal Places.
- Select Latitude Column >> Transform >> Rounding >> Round >> Decimal Places 2

= Table.TransformColumns("#Checked Distinctive Values of All Column",{{"Latitude", each Number.Round(_, 2), type number}})

The screenshot displays the Microsoft Power BI Desktop interface. The top ribbon shows the 'Transform' tab with various data manipulation options. The 'Queries' pane on the left lists 'Pharma Sales Data'. The main view shows a table with columns: 'try', '1.2 Latitude', '1.2 Longitude', 'APC Channel', 'APC Sub-channel', and 'APC Pro'. The '1.2 Latitude' column is highlighted. The formula bar at the top shows the DAX formula: `= Table.TransformColumns("#Checked Distinctive Values of All Column",{{"Latitude", each Number.Round(_, 2), type number}})`. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Rounded Off Latitude Column' as the most recent step. The status bar at the bottom indicates '18 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

try	1.2 Latitude	1.2 Longitude	APC Channel	APC Sub-channel	APC Pro
1	51.23	22.5667	Hospital	Private	Topir
2	53.42	18.4333	Pharmacy	Retail	Chori
3	50.08	18.5	Pharmacy	Institution	Acan
4	50.33	19.0833	Hospital	Private	Liolet
5	53.78	20.4942	Pharmacy	Retail	Oxym
6	54.03	22.5	Hospital	Private	Pazol
7	52.8	18.2611	Pharmacy	Retail	Symb
8	52.88	20.6106	Pharmacy	Retail	Morp
9	53.36	20.425	Hospital	Private	Lova
10	50.06	19.9372	Hospital	Private	Amp
11	51.95	14.7333	Hospital	Government	Tracc
12	52.8	21.9	Pharmacy	Institution	Atra
13	51.4	21.1567	Hospital	Private	Exotr
14	54.37	18.6333	Pharmacy	Retail	Afini
15	50.45	18.8667	Hospital	Private	Ama
16	50.15	18.8333	Hospital	Private	Neur
17	52.8	17.2	Hospital	Private	Tracc
18	51.12	15.5833	Hospital	Private	Nova
19	51.13	15.9198	Pharmacy	Institution	Paro
20	53.86	20.9569	Pharmacy	Retail	Danti
21	50.55	22.7222	Pharmacy	Institution	Arise

8. Rounded Off Longitude Column

- Rounded Off “Longitude” Column with 2 Decimal Places.
- Select Longitude Column >> Transform >> Rounding >> Round >> Decimal Places 2

= Table.TransformColumns("#Rounded Off Latitude Column",{{"Longitude", each Number.Round(_, 2), type number}})

The screenshot displays the Microsoft Power BI Desktop interface. The 'Queries' pane on the left shows 'Pharma Sales Data'. The main view is a table with columns: 'try', '1.2 Latitude', '1.2 Longitude', 'APC Channel', 'APC Sub-channel', and 'APC Proc'. The '1.2 Longitude' column is highlighted in green. The 'Formulas' bar at the top shows the DAX formula: `= Table.TransformColumns("#Rounded Off Latitude Column",{{"Longitude", each Number.Round(_, 2), type number}})`. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Rounded Off Longitude Column' at the bottom. A tooltip 'Rounded Off Longitude Column' is visible over this step. The status bar at the bottom indicates '18 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

try	1.2 Latitude	1.2 Longitude	APC Channel	APC Sub-channel	APC Proc
1	51.23	22.57	Hospital	Private	Topic
2	53.42	18.43	Pharmacy	Retail	Chon
3	50.08	18.5	Pharmacy	Institution	Acan
4	50.33	19.08	Hospital	Private	Liolet
5	53.78	20.49	Pharmacy	Retail	Oxym
6	54.03	22.5	Hospital	Private	Pazol
7	52.8	18.26	Pharmacy	Retail	Symb
8	52.88	20.61	Pharmacy	Retail	Morp
9	53.36	20.42	Hospital	Private	Lovaj
10	50.06	19.94	Hospital	Private	Amp
11	51.95	14.73	Hospital	Government	Tracc
12	52.8	21.9	Pharmacy	Institution	Atrak
13	51.4	21.16	Hospital	Private	Exotr
14	54.37	18.63	Pharmacy	Retail	Afinit
15	50.45	18.87	Hospital	Private	Amav
16	50.15	18.83	Hospital	Private	Neur
17	52.8	17.2	Hospital	Private	Tracc
18	51.12	15.58	Hospital	Private	Nova
19	51.13	15.92	Pharmacy	Institution	Paror
20	53.86	20.96	Pharmacy	Retail	Dant
21	50.55	22.72	Pharmacy	Institution	Arior

9. Renamed Column Name from Price to Unit Price, Sales to Sales Amount

- Renamed the Column Name of “Price” into “Unit Price” and “Sales” into “Sales Amount” to Access easily.

= Table.RenameColumns("#Rounded Off Longitude Column",{"Price", "Unit Price"}, {"Sales", "Sales Amount"})

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a table named 'Pharma Sales Data' with the following columns: Product Class, Quantity, Unit Price, Sales Amount, and Month. The data is organized into rows, with the first row showing 'd Stabilizers' having a quantity of 4, a unit price of 368.00, and sales amount of 1472. The right sidebar contains the 'Query Settings' pane, which lists the 'APPLIED STEPS' for the query. The first step is 'Renamed Column Name From Price to Unit Price and Sales to Sales Amount'. The formula bar at the top shows the DAX formula: = Table.RenameColumns("#Rounded Off Longitude Column", {"Price", "Unit Price"}, {"Sales", "Sales Amount"}).

Product Class	Quantity	Unit Price	Sales Amount	Month
d Stabilizers	4	368.00	1472	January
biotics	7	591.00	4137	January
biotics	30	66.00	1980	January
gesics	6	435.00	2610	January
gesics	20	458.00	9160	January
d Stabilizers	5	123.00	615	January
gesics	20	536.00	10720	January
d Stabilizers	5	742.00	3710	January
d Stabilizers	4	551.00	2204	January
gesics	10	243.00	2430	January
d Stabilizers	35	762.00	26670	January
septics	4	249.00	996	January
d Stabilizers	5	785.00	3925	January
piretics	6	286.00	1716	January
piretics	7	487.00	3409	January
gesics	5	605.00	3025	January
piretics	5	548.00	2740	January
septics	30	288.00	8640	January
gesics	2	414.00	828	January
septics	2	307.00	614	January
piretics	5	421.00	2105	January

10. Changed Data Type of “Sales Amount” into “Fixed Decimal Number”

Mini Project Power BI

FileHomeTransformAdd ColumnViewToolsHelp

Group By

Use First Row as Headers

Count Rows

Table

Transpose

Reverse Rows

Detect Data Type

Rename

Replace Values

Fill

Pivot Column

Convert to List

Unpivot Columns

Move

Split Column

Format

Extract

Parse

Merge Columns

Statistics

Standard

Scientific

Information

Trigonometry

Rounding

Date

Time

Duration

Run R script

Run Python script

Queries [1]

Pharma Sales Data

Table.TransformColumnTypes(#"Renamed Column Name From Price to Unit Price and Sales to Sales"

Product Class	Quantity	Unit Price	Sales Amount	Month
1d Stabilizers	4	368.00	1,472.00	January
2biotics	7	591.00	4,137.00	January
3biotics	30	66.00	1,980.00	January
4gesics	6	435.00	2,610.00	January
5gesics	20	458.00	9,160.00	January
6d Stabilizers	5	123.00	615.00	January
7gesics	20	536.00	10,720.00	January
8d Stabilizers	5	742.00	3,710.00	January
9d Stabilizers	4	551.00	2,204.00	January
10gesics	10	243.00	2,430.00	January
11d Stabilizers	35	762.00	26,670.00	January
12septics	4	249.00	996.00	January
13d Stabilizers	5	785.00	3,925.00	January
14piretics	6	286.00	1,716.00	January
15piretics	7	487.00	3,025.00	January
16gesics	5	605.00	2,740.00	January
17piretics	5	548.00	8,640.00	January
18septics	30	288.00	828.00	January
19gesics	2	414.00	614.00	January
20septics	2	307.00	2,105.00	January
21piretics	5	421.00	15,300.00	January
22septics	20	765.00	1,075.00	January
23biotics	5	215.00	17,914.00	January
24septics	26	689.00		

Query Settings

PROPERTIES

NamePharma Sales Data

All Properties

APPLIED STEPS

Source

Use First Row as Headers

Changed Data Type of Quanti...

Transformed Format of Sales ...

Checked Distinctive Values of ...

Rounded Off Latitude Column

Rounded Off Longitude Colu...

Renamed Column Name Fro...

Changed Data Type of Sales ...

18 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

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11. Calculated Average of Sales Amount to Check Sales Category

- Calculated Average value of “Sales Amount” to create the “Sales Category” Column using Average of Sales Amount.
- Select Specific Column >> Transform >> Statistics >> Average.

= List.Average(#"Added Conditional Column Sales Category"[Sales Amount])

The screenshot shows the Power BI Desktop interface with the 'Transform' tab selected. The formula bar contains the following DAX formula:

```
= List.Average(#"Changed Data Type of Sales Amount into Fixed Decimal Number"[Sales Amount])
```

The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes:

- Source
- Use First Row as Headers
- Changed Data Type of Quanti...
- Transformed Format of Sales ...
- Checked Distinctive Values of ...
- Rounded Off Latitude Column
- Rounded Off Longitude Colu...
- Renamed Column Name Fro...
- Changed Data Type of Sales ...
- Calculated Average of Sales A...

A tooltip is visible over the last step, 'Calculated Average of Sales A...', with the text: 'Calculated Average of Sales Amount to Check Sales Category'.

12. Added Conditional Column Sales Category

- Added a New Conditional Column “Sales Category”, based on the Average value of “Sales Amount” Column.
- Add Column >> Conditional Column >> by created “If Condition” as follows,
 - If the “Sales Amount” is greater than or equal to “46437.715552459442” Then the output is “Above Average”.
 - If Else, Then the output is “Below Average”.

= Table.AddColumn(#"Changed Data Type of Sales Amount into Fixed Decimal Number", "Sales Category", each if [Sales Amount] >= 46437.715552459442 then "Above Average" else "Below Average")

The screenshot displays the Power Query Editor interface. The 'Add Column' tab is active, showing the 'Conditional Column' option. The formula bar contains the following M code:

```
= Table.AddColumn(#"Changed Data Type of Sales Amount into Fixed Decimal Number", "Sales Category",  
each if [Sales Amount] >= 46437.715552459442 then "Above Average" else "Below Average")
```

The data table below shows the results of the query, with columns: Year, Name of Sales Rep, Manager, Sales Team, and Sales Category. The 'Sales Category' column contains values 'Below Average' for all rows shown.

	Year	Name of Sales Rep	Manager	Sales Team	Sales Category
1	2018	Mary Gerrard	Britanny Bold	DELTA	Below Average
2	2018	Jessica Smith	Britanny Bold	DELTA	Below Average
3	2018	Steve Pepple	Tracy Banks	BRAVO	Below Average
4	2018	Mary Gerrard	Britanny Bold	DELTA	Below Average
5	2018	Anne Wu	Britanny Bold	DELTA	Below Average
6	2018	Thompson Crawford	James Goodwill	ALFA	Below Average
7	2018	Sheila Stones	Britanny Bold	DELTA	Below Average
8	2018	Mary Gerrard	Britanny Bold	DELTA	Below Average
9	2018	Stella Given	Alisha Cordwell	CHARLIE	Below Average
10	2018	Morris Garcia	Tracy Banks	BRAVO	Below Average
11	2018	Erica Jones	James Goodwill	ALFA	Below Average
12	2018	Abigail Thompson	Tracy Banks	BRAVO	Below Average
13	2018	Abigail Thompson	Tracy Banks	BRAVO	Below Average
14	2018	Jessica Smith	Britanny Bold	DELTA	Below Average
15	2018	Anne Wu	Britanny Bold	DELTA	Below Average
16	2018	Morris Garcia	Tracy Banks	BRAVO	Below Average
17	2018	Daniel Gates	Alisha Cordwell	CHARLIE	Below Average
18	2018	Morris Garcia	Tracy Banks	BRAVO	Below Average
19	2018	Anne Wu	Britanny Bold	DELTA	Below Average
20	2018	Alan Ray	James Goodwill	ALFA	Below Average
21	2018	Erin Gray	Alisha Cordwell	CHARLIE	Below Average

The right-hand pane shows the 'Query Settings' for 'Pharma Sales Data'. The 'APPLIED STEPS' list includes: Source, Use First Row as Headers, Changed Data Type of Quanti..., Transformed Format of Sales..., Checked Distinctive Values of..., Rounded Off Latitude Column, Rounded Off Longitude Colu..., Renamed Column Name Fro..., Changed Data Type of Sales..., and Added Conditional Column Sales Category (highlighted).

13. Changed the Data Type of “Sales Category” into Text

= Table.TransformColumnTypes("#Added Conditional Column Sales Category",{{"Sales Category", type text}})

The screenshot displays the Power Query Editor interface. The main area shows a table with 5 columns: Year, Name of Sales Rep, Manager, Sales Team, and Sales Category. The data consists of 24 rows, each representing a sales record from 2018. The 'Sales Category' column contains the value 'Below Average' for all entries. On the right side, the 'Query Settings' pane is open, showing the 'APPLIED STEPS' list. The steps include 'Source', 'Use First Row as Headers', 'Changed Data Type of Quanti...', 'Transformed Format of Sales ...', 'Checked Distinctive Values of ...', 'Rounded Off Latitude Column', 'Rounded Off Longitude Colu...', 'Renamed Column Name Fro...', 'Changed Data Type of Sales ...', 'Added Conditional Column S...', and 'Changed Data Type of Sales C...'. The last step, 'Changed Data Type of Sales C...', is highlighted, and a tooltip appears over it with the text 'Changed Data Type of Sales Category into Text'.

	Year	Name of Sales Rep	Manager	Sales Team	Sales Category
1	2018	Mary Gerrard	Britanny Bold	DELTA	Below Average
2	2018	Jessica Smith	Britanny Bold	DELTA	Below Average
3	2018	Steve Pepple	Tracy Banks	BRAVO	Below Average
4	2018	Mary Gerrard	Britanny Bold	DELTA	Below Average
5	2018	Anne Wu	Britanny Bold	DELTA	Below Average
6	2018	Thompson Crawford	James Goodwill	ALFA	Below Average
7	2018	Sheila Stones	Britanny Bold	DELTA	Below Average
8	2018	Mary Gerrard	Britanny Bold	DELTA	Below Average
9	2018	Stella Given	Alisha Cordwell	CHARLIE	Below Average
10	2018	Morris Garcia	Tracy Banks	BRAVO	Below Average
11	2018	Erica Jones	James Goodwill	ALFA	Below Average
12	2018	Abigail Thompson	Tracy Banks	BRAVO	Below Average
13	2018	Abigail Thompson	Tracy Banks	BRAVO	Below Average
14	2018	Jessica Smith	Britanny Bold	DELTA	Below Average
15	2018	Anne Wu	Britanny Bold	DELTA	Below Average
16	2018	Morris Garcia	Tracy Banks	BRAVO	Below Average
17	2018	Daniel Gates	Alisha Cordwell	CHARLIE	Below Average
18	2018	Morris Garcia	Tracy Banks	BRAVO	Below Average
19	2018	Anne Wu	Britanny Bold	DELTA	Below Average
20	2018	Alan Ray	James Goodwill	ALFA	Below Average
21	2018	Jimmy Grey	Alisha Cordwell	CHARLIE	Below Average
22	2018	Abigail Thompson	Tracy Banks	BRAVO	Below Average
23	2018	Jessica Smith	Britanny Bold	DELTA	Below Average
24	2018	Morris Garcia	Tracy Banks	BRAVO	Below Average

14. Added Index Number from 1 for Reference

- Added Index Number from 1, to assign the unique ID for each sales for further transformations.

The screenshot displays the Microsoft Power BI Desktop interface. The top ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing various data manipulation options. The main workspace displays a table named 'Pharma Sales Data' with the following columns: Name of Sales Rep, Manager, Sales Team, Sales Category, and Index. The Index column contains sequential numbers from 1 to 24. The right sidebar shows the 'Query Settings' pane with the 'APPLIED STEPS' list, which includes 'Added Index' as the most recent step.

	Name of Sales Rep	Manager	Sales Team	Sales Category	Index
1	ry Gerrard	Britanny Bold	DELTA	Below Average	1
2	ica Smith	Britanny Bold	DELTA	Below Average	2
3	je Pepple	Tracy Banks	BRAVO	Below Average	3
4	ry Gerrard	Britanny Bold	DELTA	Below Average	4
5	je Wu	Britanny Bold	DELTA	Below Average	5
6	mpson Crawford	James Goodwill	ALFA	Below Average	6
7	ila Stones	Britanny Bold	DELTA	Below Average	7
8	ry Gerrard	Britanny Bold	DELTA	Below Average	8
9	la Given	Alisha Cordwell	CHARLIE	Below Average	9
10	rris Garcia	Tracy Banks	BRAVO	Below Average	10
11	a Jones	James Goodwill	ALFA	Below Average	11
12	gail Thompson	Tracy Banks	BRAVO	Below Average	12
13	gail Thompson	Tracy Banks	BRAVO	Below Average	13
14	ica Smith	Britanny Bold	DELTA	Below Average	14
15	je Wu	Britanny Bold	DELTA	Below Average	15
16	rris Garcia	Tracy Banks	BRAVO	Below Average	16
17	iel Gates	Alisha Cordwell	CHARLIE	Below Average	17
18	rris Garcia	Tracy Banks	BRAVO	Below Average	18
19	je Wu	Britanny Bold	DELTA	Below Average	19
20	h Ray	James Goodwill	ALFA	Below Average	20
21	my Grey	Alisha Cordwell	CHARLIE	Below Average	21
22	gail Thompson	Tracy Banks	BRAVO	Below Average	22
23	ica Smith	Britanny Bold	DELTA	Below Average	23
24	rris Garcia	Tracy Banks	BRAVO	Below Average	24

15. Renamed Index Column into “Sales Reference ID”

Queries [1]
Query Settings

= Table.RenameColumns(#"Added Index from 1 For Reference",{{"Index","Sales Reference ID"}})

Name of Sales Rep	A ^C Manager	A ^C Sales Team	A ^C Sales Category	Sales Reference ID
y Gerrard	Britanny Bold	DELTA	Below Average	1
ca Smith	Britanny Bold	DELTA	Below Average	2
e Pepple	Tracy Banks	BRAVO	Below Average	3
y Gerrard	Britanny Bold	DELTA	Below Average	4
e Wu	Britanny Bold	DELTA	Below Average	5
mnpson Crawford	James Goodwill	ALFA	Below Average	6
a Stones	Britanny Bold	DELTA	Below Average	7
y Gerrard	Britanny Bold	DELTA	Below Average	8
a Given	Alisha Cordwell	CHARLIE	Below Average	9
ris Garcia	Tracy Banks	BRAVO	Below Average	10
j Jones	James Goodwill	ALFA	Below Average	11
ail Thompson	Tracy Banks	BRAVO	Below Average	12
ail Thompson	Tracy Banks	BRAVO	Below Average	13
ca Smith	Britanny Bold	DELTA	Below Average	14
e Wu	Britanny Bold	DELTA	Below Average	15
ris Garcia	Tracy Banks	BRAVO	Below Average	16
el Gates	Alisha Cordwell	CHARLIE	Below Average	17
ris Garcia	Tracy Banks	BRAVO	Below Average	18
e Wu	Britanny Bold	DELTA	Below Average	19
Ray	James Goodwill	ALFA	Below Average	20
ny Grey	Alisha Cordwell	CHARLIE	Below Average	21
ail Thompson	Tracy Banks	BRAVO	Below Average	22
ca Smith	Britanny Bold	DELTA	Below Average	23
ris Garcia	Tracy Banks	BRAVO	Below Average	24

PROPERTIES

Name
Pharma Sales Data

All Properties

APPLIED STEPS

- Source
- Use First Row as Headers
- Changed Data Type of Quanti...
- Transformed Format of Sales ...
- Checked Distinctive Values of ...
- Rounded Off Latitude Column
- Rounded Off Longitude Colum...
- Renamed Column Name Fro...
- Changed Data Type of Sales ...
- Added Conditional Column S...
- Changed Data Type of Sales C...
- Added Index from 1 For Refer...
- Renamed Index Column as Sa...

16. Changed Data Type of “Sales Reference ID” from Number into “Text”

Queries [1]

Pharma Sales Data

Table.TransformColumnTypes(#"Renamed Index column as Sales Reference ID",{"Sales Reference ID",

	Name of Sales Rep	Manager	Sales Team	Sales Category	Sales Reference ID
1	Mary Gerrard	Britanny Bold	DELTA	Below Average	1
2	essica Smith	Britanny Bold	DELTA	Below Average	2
3	Steve Pepple	Tracy Banks	BRAVO	Below Average	3
4	Mary Gerrard	Britanny Bold	DELTA	Below Average	4
5	Anne Wu	Britanny Bold	DELTA	Below Average	5
6	Thompson Crawford	James Goodwill	ALFA	Below Average	6
7	heila Stones	Britanny Bold	DELTA	Below Average	7
8	Mary Gerrard	Britanny Bold	DELTA	Below Average	8
9	Stella Given	Alisha Cordwell	CHARLIE	Below Average	9
10	Morris Garcia	Tracy Banks	BRAVO	Below Average	10
11	Erica Jones	James Goodwill	ALFA	Below Average	11
12	Abigail Thompson	Tracy Banks	BRAVO	Below Average	12
13	Abigail Thompson	Tracy Banks	BRAVO	Below Average	13
14	essica Smith	Britanny Bold	DELTA	Below Average	14
15	Anne Wu	Britanny Bold	DELTA	Below Average	15
16	Morris Garcia	Tracy Banks	BRAVO	Below Average	16
17	Daniel Gates	Alisha Cordwell	CHARLIE	Below Average	17
18	Morris Garcia	Tracy Banks	BRAVO	Below Average	18
19	Anne Wu	Britanny Bold	DELTA	Below Average	19
20	Man Ray	James Goodwill	ALFA	Below Average	20
21	Timmy Grey	Alisha Cordwell	CHARLIE	Below Average	21
22	Abigail Thompson	Tracy Banks	BRAVO	Below Average	22
23	essica Smith	Britanny Bold	DELTA	Below Average	23
24	Morris Garcia	Tracy Banks	BRAVO	Below Average	24

Query Settings

PROPERTIES

Name
Pharma Sales Data

All Properties

APPLIED STEPS

Source

Use First Row as Headers

Changed Data Type of Quanti...

Transformed Format of Sales ...

Checked Distinctive Values of ...

Rounded Off Latitude Column

Rounded Off Longitude Colu...

Renamed Column Name Fro...

Changed Data Type of Sales ...

Added Conditional Column S...

Changed Data Type of Sales C...

Added Index from 1 For Refer...

Renamed Index Column as Sa...

Changed Data Type of Sales R...

Changed Data Type of Sales Reference ID into Text

17. Created a New Conditional Column Customer Segment to Categorize Customers

- Created a New Conditional Column “Customer Segment” to Categorize Customers based on the “Sales Amount”.
- Add Column >> Conditional Column >> by created “If Condition” as follows,
 - If the “Sales Amount” is greater than or equal to “10000000” Then the output is “Premium Customer”.
 - If the “Sales Amount” is greater than or equal to “1000000” Then the output is “Regular Customer”.
 - If Else, Then the output is “New Customer”.

Table.AddColumn("#"Changed Data Type of Sales Reference ID into Text", "Customer Segment", each if [Sales Amount] >= 10000000 then "Premium Customer" else if [Sales Amount] >= 1000000 then "Regular Customer" else "New Customer")

	Manager	Sales Team	Sales Category	Sales Reference ID	Customer Segment
1	Britanny Bold	DELTA	Below Average	1	New Customer
2	Britanny Bold	DELTA	Below Average	2	New Customer
3	Tracy Banks	BRAVO	Below Average	3	New Customer
4	Britanny Bold	DELTA	Below Average	4	New Customer
5	Britanny Bold	DELTA	Below Average	5	New Customer
6	James Goodwill	ALFA	Below Average	6	New Customer
7	Britanny Bold	DELTA	Below Average	7	New Customer
8	Britanny Bold	DELTA	Below Average	8	New Customer
9	Nisha Cordwell	CHARLIE	Below Average	9	New Customer
10	Tracy Banks	BRAVO	Below Average	10	New Customer
11	James Goodwill	ALFA	Below Average	11	New Customer
12	Tracy Banks	BRAVO	Below Average	12	New Customer
13	Tracy Banks	BRAVO	Below Average	13	New Customer
14	Britanny Bold	DELTA	Below Average	14	New Customer
15	Britanny Bold	DELTA	Below Average	15	New Customer
16	Tracy Banks	BRAVO	Below Average	16	New Customer
17	Nisha Cordwell	CHARLIE	Below Average	17	New Customer
18	Tracy Banks	BRAVO	Below Average	18	New Customer
19	Britanny Bold	DELTA	Below Average	19	New Customer
20	James Goodwill	ALFA	Below Average	20	New Customer
21	Nisha Cordwell	CHARLIE	Below Average	21	New Customer
22	Tracy Banks	BRAVO	Below Average	22	New Customer
23	Britanny Bold	DELTA	Below Average	23	New Customer
24	Tracy Banks	BRAVO	Below Average	24	New Customer
25					

18. Changed Data Type of Customer Segment Column into Text

Queries [1]

Pharma Sales Data

Query Settings

PROPERTIES

Name

Pharma Sales Data

All Properties

APPLIED STEPS

- Source
- Use First Row as Headers
- Changed Data Type of Quanti...
- Transformed Format of Sales ...
- Checked Distinctive Values of ...
- Rounded Off Latitude Column
- Rounded Off Longitude Colu...
- Renamed Column Name Fro...
- Changed Data Type of Sales ...
- Added Conditional Column S...
- Changed Data Type of Sales C...
- Added Index Number from 1 ...
- Renamed Index Column as Sa...
- Changed Data Type of Sales R...
- Created a Calculated Column ...
- Changed Data Type of Custo...

Changed Data Type of Customer Segment into Text

	Manager	Sales Team	Sales Category	Sales Reference ID	Customer Segment
1	Britanny Bold	DELTA	Below Average	1	New Customer
2	Britanny Bold	DELTA	Below Average	2	New Customer
3	Tracy Banks	BRAVO	Below Average	3	New Customer
4	Britanny Bold	DELTA	Below Average	4	New Customer
5	Britanny Bold	DELTA	Below Average	5	New Customer
6	James Goodwill	ALFA	Below Average	6	New Customer
7	Britanny Bold	DELTA	Below Average	7	New Customer
8	Britanny Bold	DELTA	Below Average	8	New Customer
9	Nisha Cordwell	CHARLIE	Below Average	9	New Customer
10	Tracy Banks	BRAVO	Below Average	10	New Customer
11	James Goodwill	ALFA	Below Average	11	New Customer
12	Tracy Banks	BRAVO	Below Average	12	New Customer
13	Tracy Banks	BRAVO	Below Average	13	New Customer
14	Britanny Bold	DELTA	Below Average	14	New Customer
15	Britanny Bold	DELTA	Below Average	15	New Customer
16	Tracy Banks	BRAVO	Below Average	16	New Customer
17	Nisha Cordwell	CHARLIE	Below Average	17	New Customer
18	Tracy Banks	BRAVO	Below Average	18	New Customer
19	Britanny Bold	DELTA	Below Average	19	New Customer
20	James Goodwill	ALFA	Below Average	20	New Customer
21	Nisha Cordwell	CHARLIE	Below Average	21	New Customer

19. Merged the City and Country Column to Create Location Column

- Merged the “City” and “Country” columns to create a new column “Location” in the format of “City, Country”.
- Add Column >> Merge Column >> Select Separator (,) >> OK.

Queries [1]

Pharma Sales Data

Table.AddColumn(#'Changed Data Type of Customer Segment into Text', 'Merged', each Text.Combine({ ... })

Sales Team	Sales Category	Sales Reference ID	Customer Segment	Merged
1 DELTA	Below Average	1	New Customer	Lublin, Poland
2 DELTA	Below Average	2	New Customer	Świecie, Poland
3 BRAVO	Below Average	3	New Customer	Rybnik, Poland
4 DELTA	Below Average	4	New Customer	Czeladź, Poland
5 DELTA	Below Average	5	New Customer	Olštyn, Poland
6 ALFA	Below Average	6	New Customer	Olecko, Poland
7 DELTA	Below Average	7	New Customer	Inowrocław, Poland
8 DELTA	Below Average	8	New Customer	Ciechanów, Poland
9 CHARLIE	Below Average	9	New Customer	Nidzica, Poland
10 BRAVO	Below Average	10	New Customer	Kraków, Poland
11 ALFA	Below Average	11	New Customer	Gubin, Poland
12 BRAVO	Below Average	12	New Customer	Ostrów Mazowiecka, Poland
13 BRAVO	Below Average	13	New Customer	Radom, Poland
14 DELTA	Below Average	14	New Customer	Gdańsk, Poland
15 DELTA	Below Average	15	New Customer	Tarnowskie Góry, Poland
16 BRAVO	Below Average	16	New Customer	Łaziska Górne, Poland
17 CHARLIE	Below Average	17	New Customer	Wągrowiec, Poland
18 BRAVO	Below Average	18	New Customer	Lwówek Śląski, Poland
19 DELTA	Below Average	19	New Customer	Złotoryja, Poland
20 ALFA	Below Average	20	New Customer	Biskupiec, Poland
21 CHARLIE	Below Average	21	New Customer	Bilgoraj, Poland
22 BRAVO	Below Average	22	New Customer	Merged City and Country Columns to Create Location Column
23 DELTA	Below Average	23	New Customer	Bolesławiec, Poland
24 BRAVO	Below Average	24	New Customer	Świdnica, Poland

Query Settings

NAME

Pharma Sales Data

APPLIED STEPS

Source

Use First Row as Headers

Changed Data Type of Quanti...

Transformed Format of Sales ...

Checked Distinctive Values of ...

Rounded Off Latitude Column

Rounded Off Longitude Colu...

Renamed Column Name Fro...

Changed Data Type of Sales ...

Added Conditional Column S...

Changed Data Type of Sales C...

Added Index Number from 1 ...

Renamed Index Column as Sa...

Changed Data Type of Sales R...

Created a Calculated Column ...

Changed Data Type of Custo...

Merged City and Country Col...

20. Renamed the Merged Column into “Location”

Queries [1]

Pharma Sales Data

fx

= Table.RenameColumns("#Merged City and Country Columns to Create Location Column",{"Merged",

	APC Sales Team	APC Sales Category	APC Sales Reference ID	APC Customer Segment	APC Location
1	DELTA	Below Average	1	New Customer	Lublin, Poland
2	DELTA	Below Average	2	New Customer	Świecie, Poland
3	BRAVO	Below Average	3	New Customer	Rybnik, Poland
4	DELTA	Below Average	4	New Customer	Czeladź, Poland
5	DELTA	Below Average	5	New Customer	Olsztyn, Poland
6	ALFA	Below Average	6	New Customer	Olecko, Poland
7	DELTA	Below Average	7	New Customer	Inowrocław, Poland
8	DELTA	Below Average	8	New Customer	Ciechanów, Poland
9	CHARLIE	Below Average	9	New Customer	Nidzica, Poland
10	BRAVO	Below Average	10	New Customer	Kraków, Poland
11	ALFA	Below Average	11	New Customer	Gubin, Poland
12	BRAVO	Below Average	12	New Customer	Ostrów Mazowiecka, Poland
13	BRAVO	Below Average	13	New Customer	Radom, Poland
14	DELTA	Below Average	14	New Customer	Gdańsk, Poland
15	DELTA	Below Average	15	New Customer	Tarnowskie Góry, Poland
16	BRAVO	Below Average	16	New Customer	Łaziska Górne, Poland
17	CHARLIE	Below Average	17	New Customer	Wągrowiec, Poland
18	BRAVO	Below Average	18	New Customer	Lwówek Śląski, Poland
19	DELTA	Below Average	19	New Customer	Złotoryja, Poland
20	ALFA	Below Average	20	New Customer	Biskupiec, Poland
21	CHARLIE	Below Average	21	New Customer	Biłgoraj, Poland
22	BRAVO	Below Average	22	New Customer	Gniew, Poland
23	DELTA	Below Average	23	New Customer	Bolesławiec, Poland
24	BRAVO	Below Average	24	New Customer	Świdnica, Poland

Query Settings

PROPERTIES

Name

Pharma Sales Data

All Properties

APPLIED STEPS

Source

Use First Row as Headers

Changed Data Type of Quanti...

Transformed Format of Sales ...

Checked Distinctive Values of ...

Rounded Off Latitude Column

Rounded Off Longitude Colu...

Renamed Column Name Fro...

Changed Data Type of Sales ...

Added Conditional Column S...

Changed Data Type of Sales C...

Added Index Number from 1 ...

Renamed Index Column as Sa...

Changed Data Type of Sales R...

Created a Calculated Column ...

Changed Data Type of Custo...

Merged City and Country Col...

Renamed the Merged Colum...

Renamed the Merged Column into Location

21. Aggregated Sales Amount of Customers from Various Distributors

- Duplicated the “Pharma Sales Data” table and named as “Total Purchase Amount of Customers from Various Distributors”.
- Aggregated the “Total Sales Amount” by “Customers from Various Distributors” using Aggregation Function.
- Transform >> Group By >> Select Advance >> Select Grouping (Distributor) >> Add Grouping (Customer Name) >> Enter New Column Name >> Select Operation (Sum) >> Select Column (Sales Amount).

Queries [2]				
Pharma Sales Data				
Total Purchase Amount...				
	APC Distributor	APC Customer Name	1.2 Total Sales by Customer Name	
1	Gottlieb-Cruickshank	Zieme, Doyle and Kunze	180941	
2	Gottlieb-Cruickshank	Feest PLC	150056	
3	Gottlieb-Cruickshank	Medhurst-Beer Pharmaceutical Limited	401955	
4	Gottlieb-Cruickshank	Barton Ltd Pharma Plc	310109	
5	Gottlieb-Cruickshank	Keeling LLC Pharmacy	322199	
6	Gottlieb-Cruickshank	Runte-Marquardt Pharmaceutical Ltd	247789	
7	Gottlieb-Cruickshank	Blick, Pacocha and Schowalter	286806	
8	Gottlieb-Cruickshank	Leuschke PLC Pharmacy	138213	
9	Gottlieb-Cruickshank	Miller-Satterfield Pharma Plc	469861	
10	Gottlieb-Cruickshank	Bashirian-Kassulke Pharma Plc	533355	
11	Gottlieb-Cruickshank	Wolff Group Pharm	195506	
12	Gottlieb-Cruickshank	Denesik, Walter and Beatty Pharma Plc	85720	
13	Gottlieb-Cruickshank	Breitenberg-Kuhn Pharmacy	101613	
14	Gottlieb-Cruickshank	McClure, Zemlak and Dibbert Pharma Plc	108066	
15	Gottlieb-Cruickshank	Block-Romaguera Pharmaceutical Limited	509586	
16	Gottlieb-Cruickshank	Hahn, Rutherford and Gislason Pharmaceutical Limited	111177	
17	Gottlieb-Cruickshank	D'Amore and Sons	156209	
18	Gottlieb-Cruickshank	Koss Ltd Pharmaceutical Limited	322614	
19	Gottlieb-Cruickshank	Fadel-West	197041	
20	Gottlieb-Cruickshank	Lindgren-Simonis Pharm	261003	
21	Gottlieb-Cruickshank	Spinka, Schowalter and Conn Pharmaceutical Limited	250439	
22	Gottlieb-Cruickshank	Will and Sons Pharma Plc	141827	
23	Gottlieb-Cruickshank	Jakubowski Inc Pharmaceutical Limited	83869	
24	Gottlieb-Cruickshank	Schaden-Hirthe Pharmaceutical Ltd	383005	
25	Gottlieb-Cruickshank	Mraz Group Pharmacy	383005	

Query Settings

PROPERTIES

Name
Total Purchase Amount of Customers from Various Distributors

APPLIED STEPS

- Use First Row as Headers
- Changed Data Type of Qu...
- Transformed Format of Sal...
- Checked Distinctive Values...
- Rounded Off Latitude Colu...
- Rounded Off Longitude C...
- Renamed Column Name F...
- Changed Data Type of Sal...
- Added Conditional Colum...
- Changed Data Type of Sal...
- Added Index Number fro...
- Renamed Index Column as...
- Changed Data Type of Sal...
- Created a New Conditiona...
- Changed Data Type of Cus...
- Merged City and Country ...
- Renamed the Merged Col...
- Aggregated Sales Amount...

22. Renamed Column into “Total Purchase Amount of Customers from Various Distributors”

Queries [2]

Pharma Sales Data

Total Purchase Amount...

fx

= Table.RenameColumns("#Aggregated Sales Amount of Customers from Various Distributors",{"Total

Customer Name

1.2 Total Purchase Amount of Customers from Various Distributo...

	Customer Name	1.2 Total Purchase Amount of Customers from Various Distributo...
1	truickshank	Zieme, Doyle and Kunze
2	truickshank	Feest PLC
3	truickshank	Medhurst-Beer Pharmaceutical Limited
4	truickshank	Barton Ltd Pharma Plc
5	truickshank	Keeling LLC Pharmacy
6	truickshank	Runte-Marquardt Pharmaceutical Ltd
7	truickshank	Blick, Pacocha and Schowalter
8	truickshank	Leuschke PLC Pharmacy
9	truickshank	Miller-Satterfield Pharma Plc
10	truickshank	Bashirian-Kassulke Pharma Plc
11	truickshank	Wolff Group Pharm
12	truickshank	Denesik, Walter and Beatty Pharma Plc
13	truickshank	Breitenberg-Kuhn Pharmacy
14	truickshank	McClure, Zemlak and Dibbert Pharma Plc
15	truickshank	Block-Romaguera Pharmaceutical Limited
16	truickshank	Hahn, Rutherford and Gislason Pharmaceutical Limited
17	truickshank	D'Amore and Sons
18	truickshank	Koss Ltd Pharmaceutical Limited
19	truickshank	Fadel-West
20	truickshank	Lindgren-Simonis Pharm
21	truickshank	Spinka, Schowalter and Conn Pharmaceutical Limited
22	truickshank	Will and Sons Pharma Plc
23	truickshank	Jakubowski Inc Pharmaceutical Limited
24	truickshank	Schaden-Hirthe Pharmaceutical Ltd
25		

Renamed Column into Total Purchase Amount of Customers from Various Distributors

Query Settings

PROPERTIES

Name

Total Purchase Amount of Customers from

All Properties

APPLIED STEPS

Use First Row as Headers

Changed Data Type of Qu...

Transformed Format of Sal...

Checked Distinctive Values...

Rounded Off Latitude Colu...

Rounded Off Longitude C...

Renamed Column Name F...

Changed Data Type of Sal...

Added Conditional Colum...

Changed Data Type of Sal...

Added Index Number fro...

Renamed Index Column as...

Changed Data Type of Sal...

Created a New Conditiona...

Changed Data Type of Cus...

Merged City and Country ...

Renamed the Merged Col...

Aggregated Sales Amount...

Renamed Column into Tot...

Changed Type

23. Changed Data Type into “Fixed Decimal Number”

Queries [2]

- Pharma Sales Data
- Total Purchase Amount...

A/C Distributor	A/C Customer Name	Total Purchase Amount of Customers from Various Distributors
1 Koss	Abernathy Group	57,
2 Gleason	Abernathy Group	
3 Lockman	Abernathy Group	2,
4 Crist Inc	Abernathy Group	
5 Welch-Langworth	Abernathy Group	5,
6 Lesch	Abernathy Group	
7 Nader-Gaylord	Abernathy Group	
8 Kozey-Emmerich	Abernathy Group	21,
9 Rohan and Sons	Abernathy Group	
10 Gerlach LLC	Abernathy Group	88,
11 Bashirian-Kassulke	Abernathy Group	14,
12 Daugherty-Rempel	Abernathy Group	24,
13 Romaguera-Fay	Abernathy Group	
14 Erdman	Abernathy Group	32,
15 Carter-Conn	Abernathy Group Pharmaceutical Limited	
16 Prohaska-Kuhic	Abernathy Group Pharmaceutical Limited	2,
17 Rohan	Abernathy Group Pharmaceutical Limited	3,
18 Stehr-Champlin	Abernathy Group Pharmaceutical Limited	5,
19 Beier	Abernathy Group Pharmaceutical Limited	4,
20 Smith Inc	Abernathy Group Pharmaceutical Limited	1,
21 Cassin	Abernathy Group Pharmaceutical Limited	1,
22 Rogahn-Klein	Abernathy Group Pharmaceutical Limited	
23 Lindgren-Simonis Pharm	Abernathy Group Pharmaceutical Limited	
24 Gottlieb-Cruickshank	Abernathy Group Pharmaceutical Limited	

Query Settings

PROPERTIES

Name
Total Purchase Amount of Customers from Various Distributors

APPLIED STEPS

- Changed Data Type of Query
- Transformed Format of Sales Amount
- Checked Distinctive Values
- Rounded Off Latitude Column
- Rounded Off Longitude Column
- Renamed Column Name From A/C to Customer Name
- Changed Data Type of Sales Amount to Fixed Decimal Number
- Added Conditional Column
- Changed Data Type of Sales Amount to Fixed Decimal Number
- Added Index Number from A/C to Customer Name
- Renamed Index Column as A/C to Customer Name
- Changed Data Type of Sales Amount to Fixed Decimal Number
- Created a New Condition
- Changed Data Type of Customer Name to Fixed Decimal Number
- Merged City and Country Columns
- Renamed the Merged Column as Customer Name
- Aggregated Sales Amount by Customer Name
- Renamed Column into Total Purchase Amount
- Changed Data Type into Fixed Decimal Number

Sorted in Ascending Order to Check all Customers Purchase Amount

25. Grouped Sales Amount by Product Name to Calculate Total Sales Amount by Product Name

- Duplicated the “Pharma Sales Data” table and named as “Total Sales Amount by Product Name”.
- Transform >> Group By >> Select the Specific Column (Product Name) >> Select Operation (Sum) >> Select Column (Sales Amount).
- Changed Data Type into “Fixed Decimal Number”.

Queries [3] = Table.TransformColumnTypes(#"Grouped Sales Amount by Product Name",{{"Total Sales Amount by Product Name"}}

	Product Name	Total Sales Amount by Product Name
1	Topipizole	3,58,59,024.00
2	Choriotrisin	6,94,71,436.00
3	Acantaine	71,81,328.00
4	Lioletine Refliruvax	4,84,08,105.00
5	Oxymotroban Fexoformin	5,77,84,486.00
6	Pazofenac	1,50,22,482.00
7	Symbitrim	8,36,13,856.00
8	Morphizolid Tianalin	8,46,69,620.00
9	Lovapur	5,59,47,989.00
10	Ampysin	2,42,72,541.00
11	Trazobalamin	8,29,90,190.00
12	Atrabacin Alkerotec	2,93,49,879.00
13	Exotropin Empizine	6,63,74,105.00
14	Afinitasol	3,43,94,360.00
15	Amavirase	8,62,94,452.00
16	Neuropogen Empibax	9,17,51,275.00
17	Tracdomide	6,85,14,796.00
18	Novakyn Tracprox	3,08,08,800.00
19	Paronium Atracustone	4,03,16,976.00
20	Dantocept Ferurenone	5,32,61,430.00
21	Aciprex	5,01,44,889.00
22	Diaxolol	5,90,41,170.00
23	Saprodaine Zanprosyn	2,21,15,330.00
24	Magnemunex	8,59,25,190.00
25	Pitodomide	32,21,208.00

Query Settings

PROPERTIES

Name
Total Sales Amount by Product Name

All Properties

APPLIED STEPS

- Source
- Use First Row as Headers
- Changed Data Type of Qu...
- Transformed Format of Sal...
- Checked Distinctive Values...
- Rounded Off Latitude Colu...
- Rounded Off Longitude C...
- Renamed Column Name F...
- Changed Data Type of Sal...
- Added Conditional Colum...
- Changed Data Type of Sal...
- Added Index Number fro...
- Renamed Index Column as...
- Changed Data Type of Sal...
- Created a New Conditiona...
- Changed Data Type of Cus...
- Merged City and Country ...
- Renamed the Merged Col...
- Grouped Sales Amount by ...
- Changed Data Type into Fi...

Grouped Sales Amount by Product Name

26. Grouped Total Number of Sales by Location

- Duplicated the “Pharma Sales Data” table and named as “Total Number of Sales by Location”.
- Transform >> Group By >> Select the Specific Column (Location) >> Select Operation (Count Distinctive Rows).

The screenshot displays a data transformation tool interface. On the left, a list of queries includes "Pharma Sales Data", "Total Purchase Amount...", "Total Sales Amount by P...", and "Total Number of Sales b...". The main workspace shows a table with 25 rows of data, including columns for "Location" and "Total Number of Sales by Location". The formula bar at the top indicates a table rename operation: `Table.RenameColumns("#Grouped Number of Sales by Location",{"Number of Sales by Location", "Total" ...`. On the right, the "Query Settings" panel is open, showing the "APPLIED STEPS" list. The steps include "Source", "Use First Row as Headers", "Changed Data Type of Qu...", "Transformed Format of Sal...", "Checked Distinctive Values...", "Rounded Off Latitude Colu...", "Rounded Off Longitude C...", "Renamed Column Name F...", "Changed Data Type of Sal...", "Added Conditional Colum...", "Changed Data Type of Sal...", "Added Index Number fro...", "Renamed Index Column as...", "Changed Data Type of Sal...", "Created a New Conditiona...", "Changed Data Type of Cus...", "Merged City and Country ...", "Renamed the Merged Col...", "Grouped Number of Sales ...", and "Renamed into Total Numb...". A tooltip at the bottom right indicates the final step: "Renamed into Total Number of Sales by Location".

	Location	Total Number of Sales by Location
1	Lublin, Poland	196
2	Świecie, Poland	183
3	Rybnik, Poland	157
4	Czeladź, Poland	191
5	Olsztyn, Poland	184
6	Olecko, Poland	201
7	Inowrocław, Poland	225
8	Ciechanów, Poland	185
9	Nidzica, Poland	210
10	Kraków, Poland	214
11	Gubin, Poland	202
12	Ostrów Mazowiecka, Poland	210
13	Radom, Poland	208
14	Gdańsk, Poland	189
15	Tarnowskie Góry, Poland	217
16	Łaziska Górne, Poland	200
17	Wągrowiec, Poland	209
18	Lwówek Śląski, Poland	202
19	Złotoryja, Poland	172
20	Biskupiec, Poland	211
21	Bilgoraj, Poland	193
22	Gniew, Poland	200
23	Bolesławiec, Poland	226
24	Świdnica, Poland	205
25	Piastów, Poland	212

27. Aggregated Total Sales Amount by Month and Year

- Duplicated the “Pharma Sales Data” table and named as “Total Sales Amount by Month and Year”.
- Transform >> Group By >> Select Advance >> Select Grouping (Year) >> Add Grouping (Month) >> Enter New Column Name >> Select Operation (Sum) >> Select Column (Sales Amount).
- Changed Data Type into “Fixed Decimal Number”.

The screenshot displays the Power BI Power Query Editor interface. On the left, the 'Queries' pane lists several queries, with 'Total Sales Amount by...' selected. The main area shows a table with columns for Year, Month, and Total Sales Amount by Month and Year. The table contains 25 rows of data, spanning from 2017 to 2019. The 'Total Sales Amount by Month and Year' column is highlighted in green. On the right, the 'Query Settings' pane is open, showing the 'APPLIED STEPS' list. The last step, 'Changed Data Type into Fixed Decimal Number', is highlighted. The status bar at the bottom indicates '3 COLUMNS, 48 ROWS' and 'Column profiling based on top 1000 rows'.

	Year	Month	Total Sales Amount by Month and Year
1	2018	January	25,39,30,252.00
2	2018	February	27,50,72,609.00
3	2018	March	38,37,19,035.00
4	2018	April	24,64,10,406.00
5	2018	May	23,44,52,510.00
6	2018	June	36,39,98,450.00
7	2018	July	30,69,98,901.00
8	2018	August	27,81,69,723.00
9	2018	September	29,25,90,631.00
10	2018	October	35,98,35,083.00
11	2018	November	29,68,37,587.00
12	2018	December	21,48,82,167.00
13	2017	January	15,18,72,184.00
14	2017	February	20,89,95,889.00
15	2017	March	21,47,67,956.00
16	2017	April	18,88,00,584.00
17	2017	May	22,36,34,799.00
18	2017	June	26,66,68,391.00
19	2017	July	26,04,88,776.00
20	2017	August	23,38,34,675.00
21	2017	September	26,02,51,610.00
22	2017	October	21,93,38,825.00
23	2017	November	25,20,67,510.00
24	2017	December	22,07,59,542.00
25	2019	January	9,76,64,076.00

- All transformations completed by using Power BI Power Query Editor.
- Close and Apply to load the data into Power BI.

DAX and Visualization

- Loaded the Tables from Power Query Editor by using Close & Apply Button.
- Formatted “Unit Price” and “Total Sales Amount” Columns from Currency into “Decimal Number”.
- Checked Data type of all Columns.
- Checked Summarization status for all Numerical Columns.

DAX (Data Analysis Expressions)

1. Created a New Calculated Column Channel Type in Pharma Sales Data

- Created a New Calculated Column in Pharma Sales Data Table.
- Combined the “Channel” and “Sub-Channel” Columns into a Single Column “Channel Type”, by using following DAX Formula

Channel Type = 'Pharma Sales Data'[Channel] & " " & 'Pharma Sales Data'[Sub-channel]

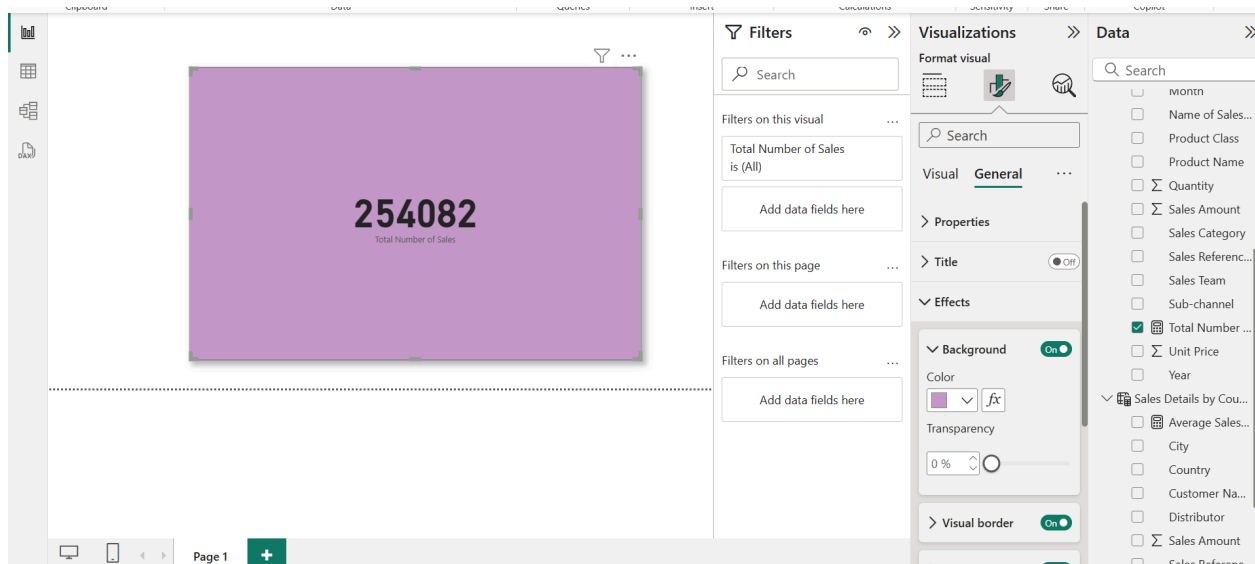
The screenshot displays the Microsoft Power BI interface. At the top, the 'Column Tools' ribbon is active, showing the 'Channel Type' column's properties: Name, Data type (Text), Format (Text), Summarization (Don't summarize), Data category (Uncategorized), and various icons for sorting, grouping, and managing relationships. Below the ribbon, the 'Data' pane on the right shows the 'Pharma Sales Data' table with the 'Channel Type' column highlighted. The main table view shows 20 rows of data with columns: Name of Sales Rep, Manager, Sales Team, Sales Category, Sales Reference ID, Customer Segment, Location, and Channel Type. The 'Channel Type' column contains the concatenated values of 'Channel' and 'Sub-channel' for each row.

Name of Sales Rep	Manager	Sales Team	Sales Category	Sales Reference ID	Customer Segment	Location	Channel Type
Yvonne Wu	Britanny Bold	DELTA	Below Average	41159	New Customer	Geislingen an der Steige, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	41851	New Customer	Lüneburg, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	45535	New Customer	Springe, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	47999	New Customer	Heidenheim, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	49365	New Customer	Achern, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	49378	New Customer	Bensheim, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	55160	New Customer	Weißenfels, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	56266	New Customer	Bad Kissingen, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	57337	New Customer	Schleswig, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	60881	New Customer	Bad Kissingen, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	62422	New Customer	Unterschleißheim, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	64370	New Customer	Herrenberg, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	68175	New Customer	Herzogenrath, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	69208	New Customer	Weingarten, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	71507	New Customer	Wolfenbüttel, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	73843	New Customer	Northeim, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	75659	New Customer	Schwandorf, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	76880	New Customer	Kevelaer, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	82937	New Customer	Leipzig, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	91829	New Customer	Herzogenrath, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	92844	New Customer	Hof, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	99412	New Customer	Pulheim, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	99644	New Customer	Meerbusch, Germany	Pharmacy Retail
Yvonne Wu	Britanny Bold	DELTA	Below Average	101702	New Customer	Friedrichshafen, Germany	Pharmacy Retail

2. Calculated Measures - Total Number of Sales

- Created a Calculated measure to Calculate Sales Count in the “Pharma Sales Data” by using following DAX Formula,

Total Number of Sales = COUNTROWS('Pharma Sales Data')

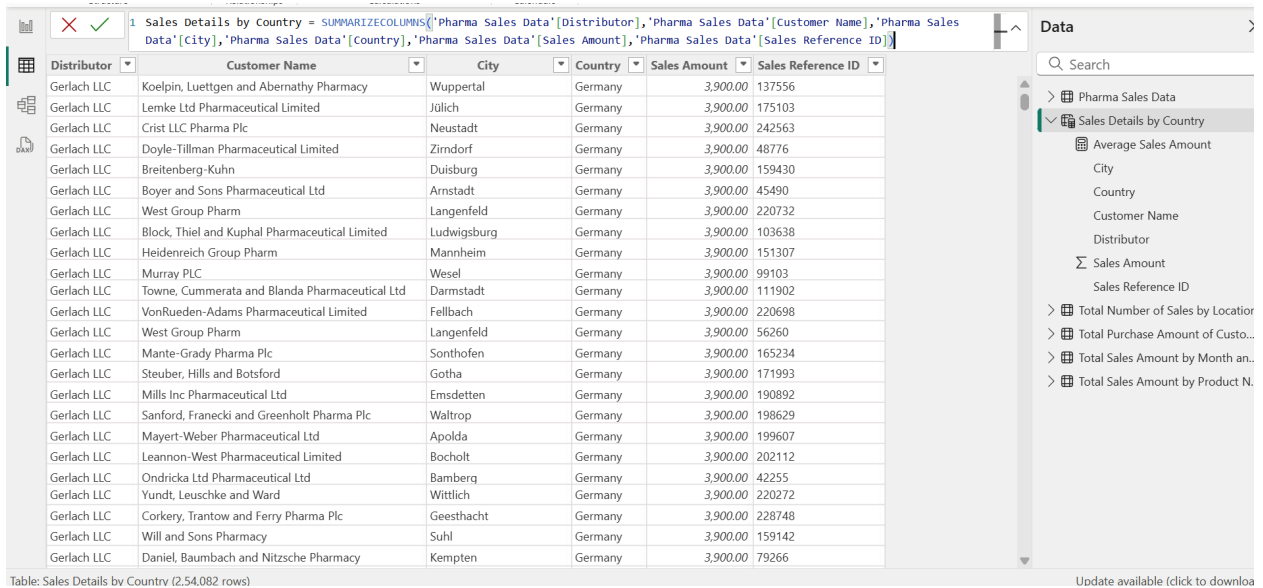


3. Created Calculated Table “Sales Details by Country”

- Created a New Calculated Table “Sales Details by Country” by using following DAX formula

Sales Details by Country = SUMMARIZECOLUMNS('Pharma Sales Data'[Distributor],'Pharma Sales Data'[Customer Name],'Pharma Sales Data'[City],'Pharma Sales Data'[Country],'Pharma Sales Data'[Sales Amount],'Pharma Sales Data'[Sales Reference ID])

- Formatted Sales Amount into “Decimal Number”.



1 Sales Details by Country = SUMMARIZECOLUMNS('Pharma Sales Data'[Distributor],'Pharma Sales Data'[Customer Name],'Pharma Sales Data'[City],'Pharma Sales Data'[Country],'Pharma Sales Data'[Sales Amount],'Pharma Sales Data'[Sales Reference ID])

Distributor	Customer Name	City	Country	Sales Amount	Sales Reference ID
Gerlach LLC	Koelpin, Luetngen and Abernathy Pharmacy	Wuppertal	Germany	3,900.00	137556
Gerlach LLC	Lemke Ltd Pharmaceutical Limited	Jülich	Germany	3,900.00	175103
Gerlach LLC	Crist LLC Pharma Plc	Neustadt	Germany	3,900.00	242563
Gerlach LLC	Doyle-Tillman Pharmaceutical Limited	Zirndorf	Germany	3,900.00	48776
Gerlach LLC	Breitenberg-Kuhn	Duisburg	Germany	3,900.00	159430
Gerlach LLC	Boyer and Sons Pharmaceutical Ltd	Arnstadt	Germany	3,900.00	45490
Gerlach LLC	West Group Pharm	Langenfeld	Germany	3,900.00	220732
Gerlach LLC	Block, Thiel and Kuphal Pharmaceutical Limited	Ludwigsburg	Germany	3,900.00	103638
Gerlach LLC	Heidenreich Group Pharm	Mannheim	Germany	3,900.00	151307
Gerlach LLC	Murray PLC	Wesel	Germany	3,900.00	99103
Gerlach LLC	Towne, Cummerata and Blanda Pharmaceutical Ltd	Darmstadt	Germany	3,900.00	111902
Gerlach LLC	VonRueden-Adams Pharmaceutical Limited	Fellbach	Germany	3,900.00	220698
Gerlach LLC	West Group Pharm	Langenfeld	Germany	3,900.00	56260
Gerlach LLC	Mante-Grady Pharma Plc	Sonthofen	Germany	3,900.00	165234
Gerlach LLC	Steuber, Hills and Botsford	Gotha	Germany	3,900.00	171993
Gerlach LLC	Mills Inc Pharmaceutical Ltd	Emsdetten	Germany	3,900.00	190892
Gerlach LLC	Sanford, Franecki and Greenholt Pharma Plc	Waltrop	Germany	3,900.00	198629
Gerlach LLC	Mayert-Weber Pharmaceutical Ltd	Apolda	Germany	3,900.00	199607
Gerlach LLC	Leannon-West Pharmaceutical Limited	Bocholt	Germany	3,900.00	202112
Gerlach LLC	Ondricka Ltd Pharmaceutical Ltd	Bamberg	Germany	3,900.00	42255
Gerlach LLC	Yundt, Leuschke and Ward	Wittlich	Germany	3,900.00	220272
Gerlach LLC	Corkery, Trantow and Ferry Pharma Plc	Geesthacht	Germany	3,900.00	228748
Gerlach LLC	Will and Sons Pharmacy	Suhl	Germany	3,900.00	159142
Gerlach LLC	Daniel, Baumbach and Nitzsche Pharmacy	Kempton	Germany	3,900.00	79266

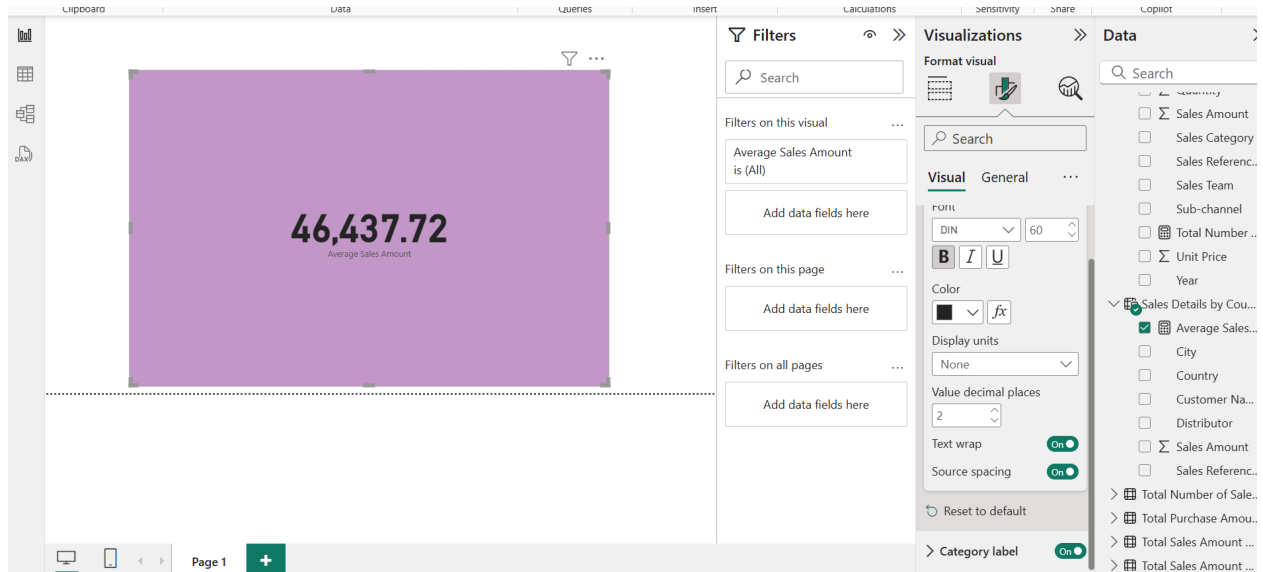
Table: Sales Details by Country (2,54,082 rows)

Update available (click to download)

4. Calculated Measure - Average Sales Amount

- Created a Calculated Measure to Calculate “Average Sales Amount” by using following DAX Formula

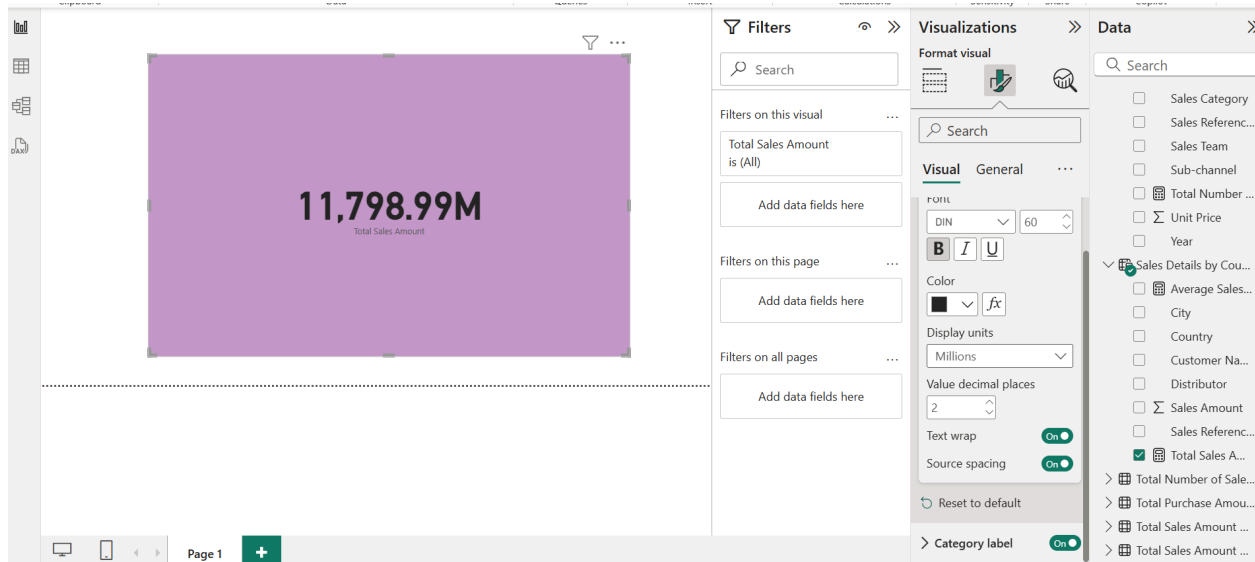
Average Sales Amount = AVERAGE('Sales Details by Country'[Sales Amount])



5. Calculated Measure - Total Sales Amount

- Created a Calculated Measure to Calculate “Total Sales Amount” by using following DAX Formula

Total Sales Amount = SUM('Sales Details by Country'[Sales Amount])



6. Created a New Calculated Column Sales Team Details

- Created a New Calculated Column “Sales Team Details” by using following DAX Formula,

Sales Team Details = SUMMARIZECOLUMNS('Pharma Sales Data'[Name of Sales Rep],'Pharma Sales Data'[Manager],'Pharma Sales Data'[Sales Team],'Pharma Sales Data'[Month],'Pharma Sales Data'[Year],'Pharma Sales Data'[Sales Amount],'Pharma Sales Data'[Sales Reference ID])

- Formatted the Sales Amount Column from Currency into Decimal Number.

The screenshot shows the Microsoft Power BI Desktop interface. At the top, the 'Formulas' ribbon is active, displaying the DAX formula for the calculated column 'Sales Team Details'. Below the formula bar, a table view shows the results of the calculation. The table has columns for 'Name of Sales Rep', 'Manager', 'Sales Team', 'Month', 'Year', 'Sales Amount', and 'Sales Reference ID'. The 'Sales Amount' column is formatted as a decimal number. On the right side, the 'Data' pane shows the hierarchy of the data model, with 'Sales Team Details' expanded to show its columns.

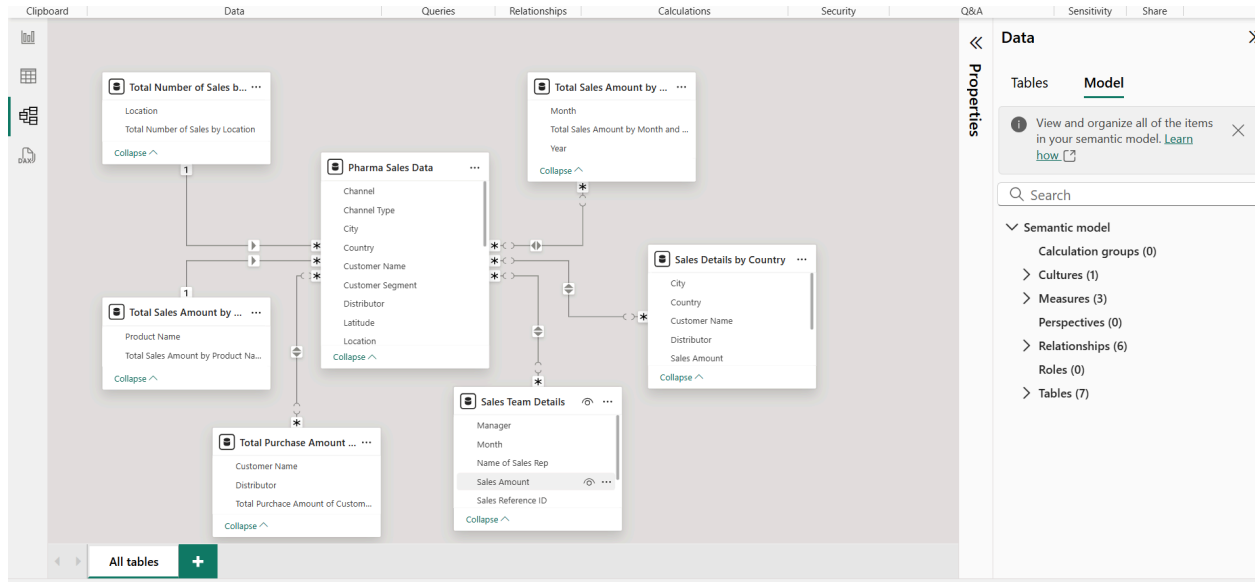
1 Sales Team Details = SUMMARIZECOLUMNS('Pharma Sales Data'[Name of Sales Rep],'Pharma Sales Data'[Manager],'Pharma Sales Data'[Sales Team],'Pharma Sales Data'[Month],'Pharma Sales Data'[Year],'Pharma Sales Data'[Sales Amount],'Pharma Sales Data'[Sales Reference ID])

Name of Sales Rep	Manager	Sales Team	Month	Year	Sales Amount	Sales Reference ID
Jessica Smith	Britanny Bold	DELTA	October	2018	5,130.00	139989
Jessica Smith	Britanny Bold	DELTA	October	2018	7,420.00	137440
Jessica Smith	Britanny Bold	DELTA	October	2018	1,540.00	140394
Jessica Smith	Britanny Bold	DELTA	October	2018	4,040.00	139198
Jessica Smith	Britanny Bold	DELTA	October	2018	7,080.00	139913
Jessica Smith	Britanny Bold	DELTA	October	2018	3,610.00	141000
Jessica Smith	Britanny Bold	DELTA	October	2018	3,470.00	137852
Jessica Smith	Britanny Bold	DELTA	October	2018	790.00	138293
Jessica Smith	Britanny Bold	DELTA	October	2018	5,630.00	138558
Jessica Smith	Britanny Bold	DELTA	October	2018	4,160.00	139948
Jessica Smith	Britanny Bold	DELTA	October	2018	2,840.00	140959
Jessica Smith	Britanny Bold	DELTA	October	2018	2,840.00	141149
Jessica Smith	Britanny Bold	DELTA	October	2018	220.00	138152
Jessica Smith	Britanny Bold	DELTA	October	2018	6,550.00	139858
Jessica Smith	Britanny Bold	DELTA	October	2018	250.00	140917
Jessica Smith	Britanny Bold	DELTA	October	2018	2,980.00	137837
Jessica Smith	Britanny Bold	DELTA	October	2018	1,910.00	137846
Jessica Smith	Britanny Bold	DELTA	October	2018	1,910.00	138348
Jessica Smith	Britanny Bold	DELTA	October	2018	3,010.00	140683
Jessica Smith	Britanny Bold	DELTA	October	2018	1,280.00	136950
Jessica Smith	Britanny Bold	DELTA	October	2018	5,900.00	137798
Jessica Smith	Britanny Bold	DELTA	October	2018	5,480.00	138137
Jessica Smith	Britanny Bold	DELTA	October	2018	6,280.00	138816

Table: Sales Team Details (2,54,082 rows) Column: Sales Amount (20,542 distinct values)

Update available (click to download)

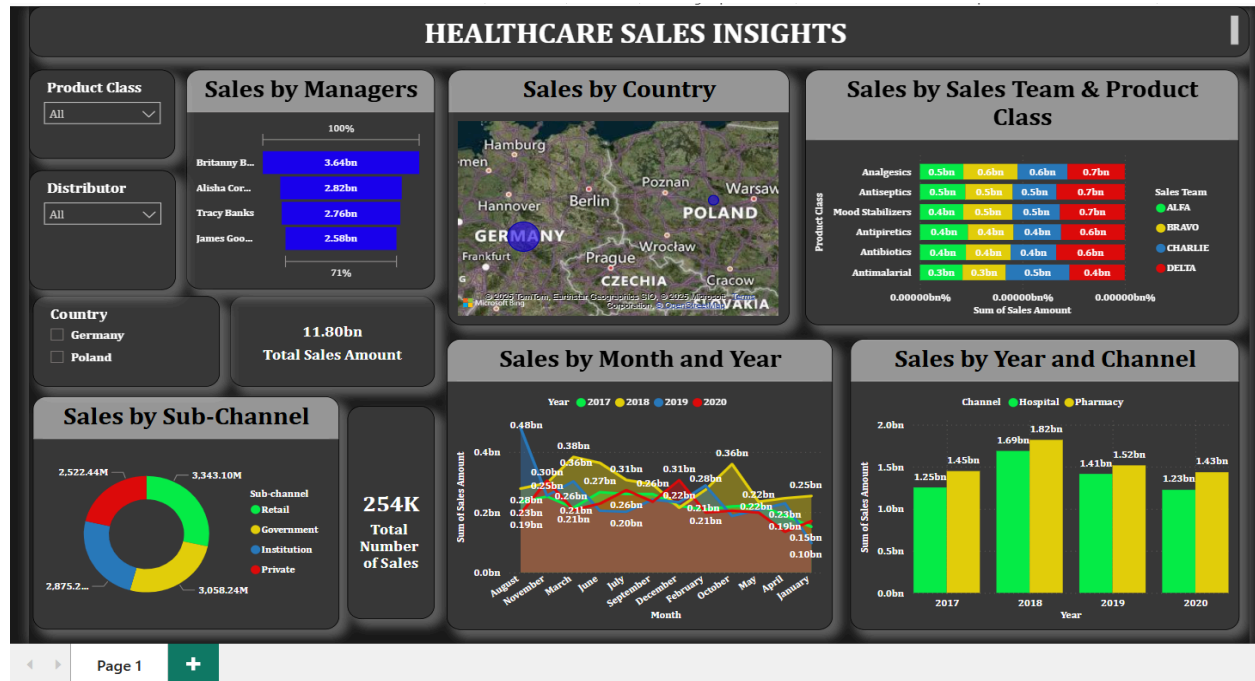
Data Modeling



Data Visualization

- View >> Select Theme >> Custom Theme
- Insert Text box to Create Title of Report “HEALTHCARE SALES INSIGHTS”

HEALTHCARE SALES INSIGHTS REPORT



Power BI Mini Project Report View - pdf

<https://drive.google.com/file/d/1AI8yGWRCzNLtSVFHVyCnovUE6fHtRDLD/view?usp=sharing>

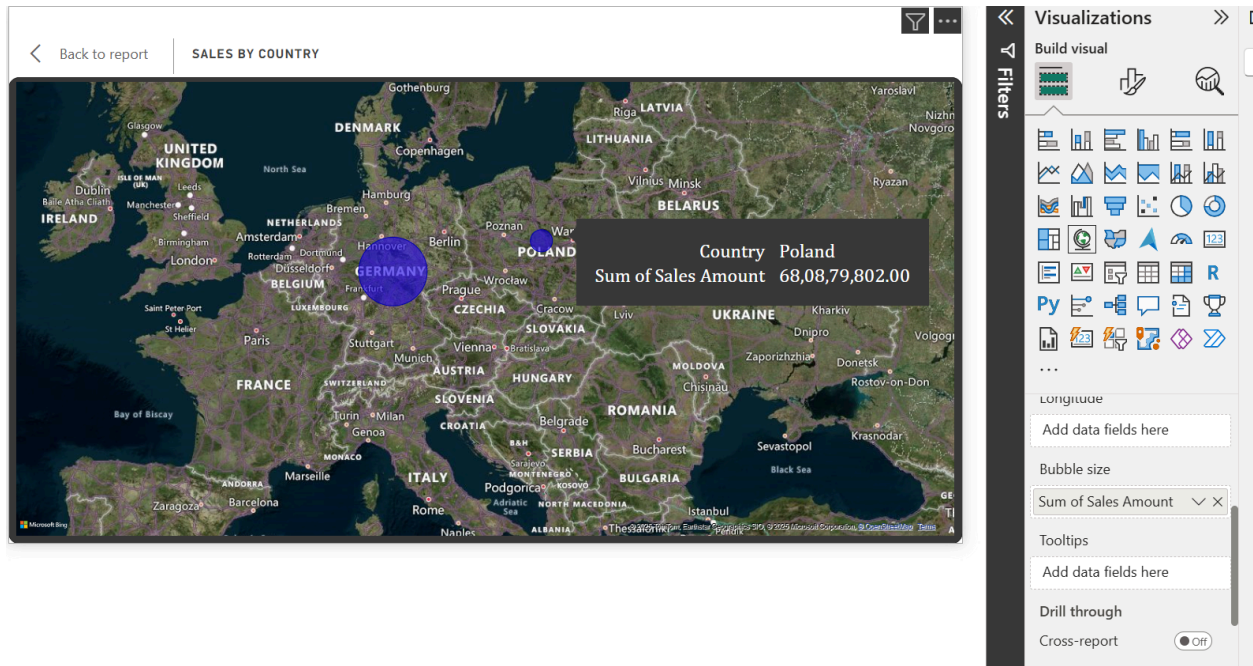
1. Sales by Year and Channel:

- Compare the Total Sales by two different Channels, Hospital and Pharmacy and by Every Year (2017,2018,2019,2020) using Clustered Column Chart.
- X-Axis = Year
- Y-Axis = Sum of Sales Amount
- Legend = Channel



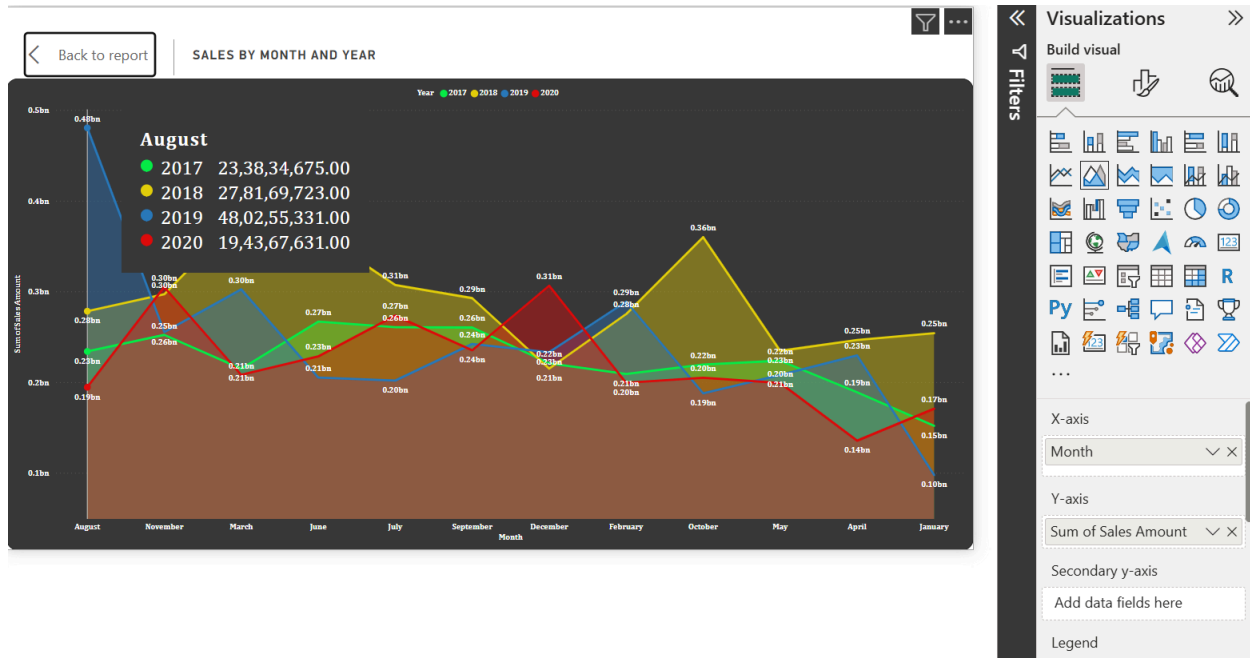
2. Sales by Country

- Visualize Total Sales on a Map by Country to Identify Regional Sales Patterns.
- Location - Country
- Bubble Size - Sum of Sales Amount



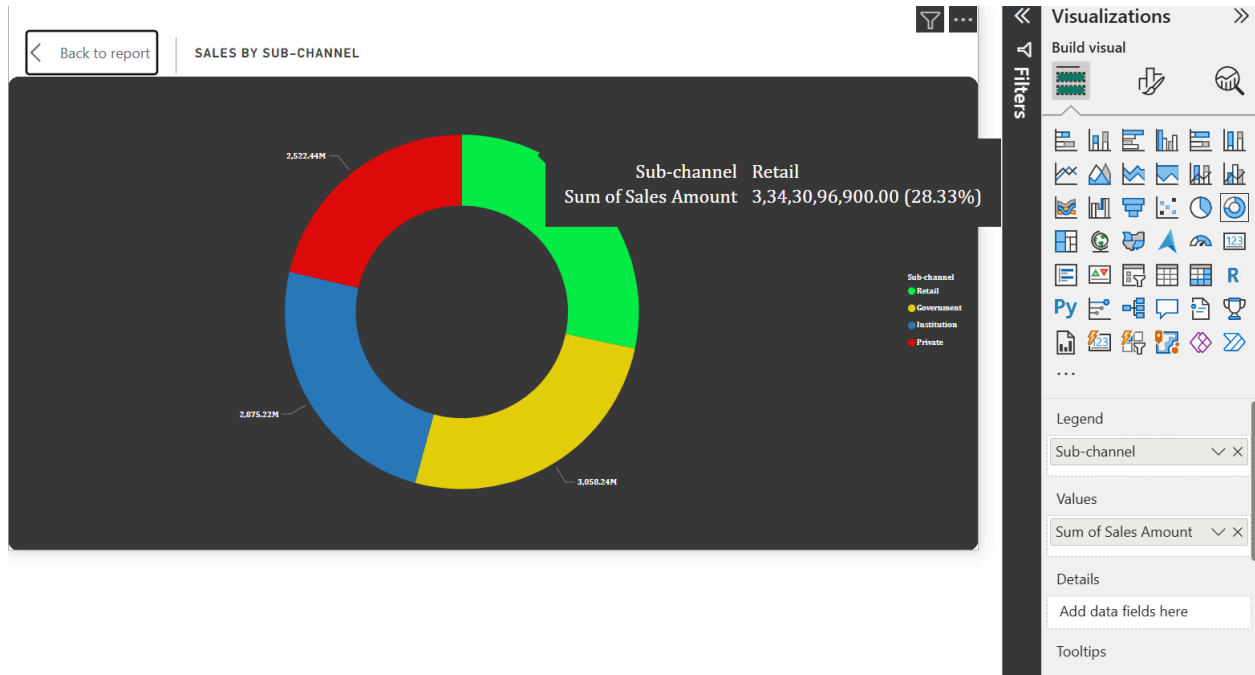
3. Sales by Month and Year

- Show the Total Sales by Month and Year using Area Chart.
- X-Axis = Month
- Y-Axis = Sum of Sales Amount
- Legend = Year



4. Sales by Sub-Channel

- Analyze the Total Sales by each Sub-channel of Customers using a Donut Chart.
- Legend = Sub-channel
- Values = Sum of Sales Amount



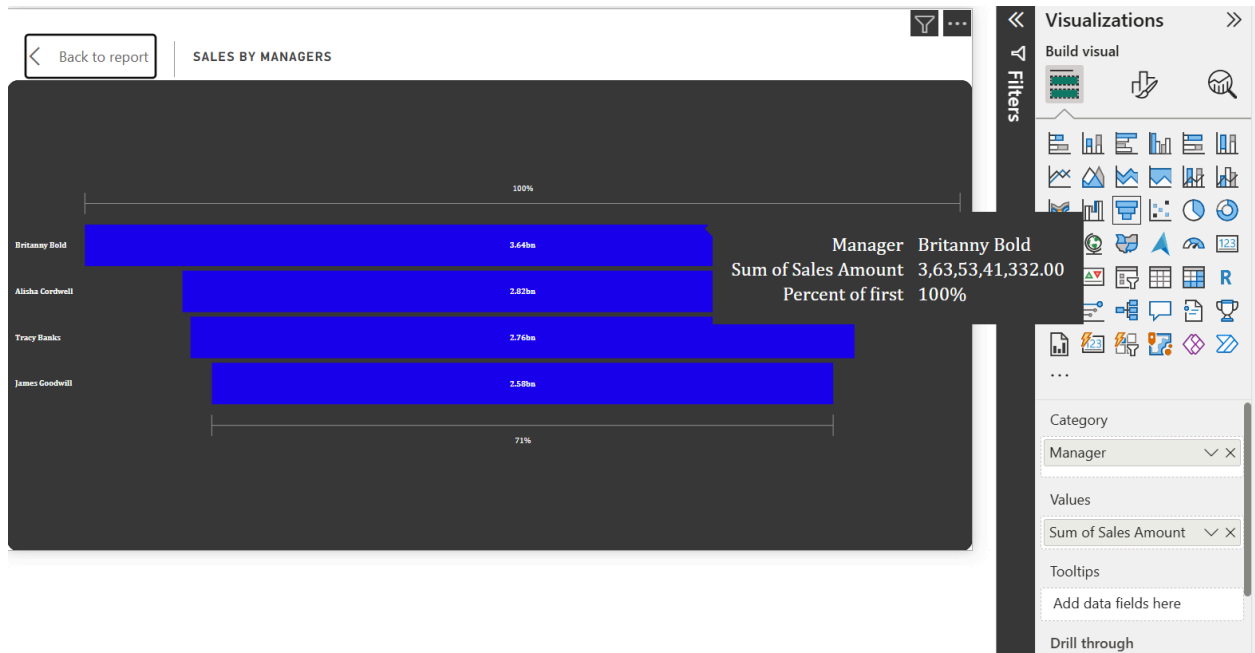
5. Sales by Sales Team and Product Class

- Analyze the overall Sales done by Every Sales Team in All Product Class using 100% Stacked Bar Chart.
- X-Axis = Sum of Sales Amount
- Y-Axis = Product Class
- Legend = Sales Team



6. Sales by Managers

- Analyze Total Sales done by Managers to Identify Top Contributors in Sales using Funnel Chart.
- Category = Manager
- Values = Sum of Sales Amount



7.Cards - Total Sales Amount

- Created Cards to Display the Total Sales Amount.
- For Quick Comparison and Analysis.

8. Cards - Number of Sales

- Created a Cards to Display the Total Number of Sales.
- For Quick Comparison and Analysis.

9. Slicer - Country

- Created a Slicer - Country.
- Slicer connected with all Visuals to slice the data dynamically across all visuals by Country.
- Helps in quick and flexible data exploration.

10. Slicer - Distributors

- Created a Slicer - Distributors.
- Slicer connected with all Visuals to slice the data dynamically across all visuals by Various Distributors.
- Helps in quick and flexible data exploration.

11. Slicer - Product Class

- Created a Slicer - Product Class.
- Slicer connected with all Visuals to slice the data dynamically across all visuals by Various Product Class.
- Helps in quick and flexible data exploration.